

Technetium

43
Tc

The atoms of this highly radioactive element are very unstable and break down quickly. Technetium was named after “teknetos”, the Greek word for “artificial”. It is the first artificial element to have been created.

ATOMIC MASS (98)

STATE Solid

DISCOVERY 1937 (Carlo Perrier and Emilio Segrè)



Foil
of pure
technetium

Ruthenium

44
Ru

Ruthenia, the Latin name for Russia, inspired this rare element's name. Ruthenium is used in turning coal into the liquid fuel petroleum, in X-ray machines, and in making solar cells (devices that turn sunlight into electricity).



Laboratory-
refined
crystals of
pure ruthenium

ATOMIC MASS 101.07

STATE Solid

DISCOVERY 1808 (Jedrzej Sniadecki), 1825 (Gottfried Osann), 1844 (Karl Karlovich Klaus)

Palladium

46
Pd

This precious metal is added in small amounts to gold to make white-gold jewellery. A compound called palladium chloride is an ingredient in sensors that detect levels of poisonous carbon monoxide gas.

ATOMIC MASS

106.42

STATE Solid

DISCOVERY

1803 (William Hyde Wollaston)



Piece of
pure palladium